

Motivic Homotopy

Naren

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This note goes over the basic notions of schemes and gives a minimal working knowledge of motivic homotopy. I have just sourced stuff here mainly for my reference.

1 Basic algebraic geometry

In algebraic geometry $\text{Spec}(X) = (\text{Spec}(X), \mathcal{O}_{\text{Spec}(X)})$ where $\text{Spec}(X)$ is a topological space and $\mathcal{O}$ is a sheaf of rings. We have the isomorphism of hom sets $\text{hom}(\text{Spec}(R), X) \cong \text{hom}(\mathcal{O}_X, R)$. The functor of points view:

$$\begin{aligned} \text{Spec}(X) : \text{Crings} &\rightarrow \text{Set} \\ R &\mapsto \text{hom}(X, R) \end{aligned}$$

$$\begin{aligned} \text{Rings}^{op} &\xrightarrow{\cong} \text{Category of schemes} \\ R &\mapsto \text{Spec}(R) \\ \mathcal{O}_X &\leftarrow X \end{aligned}$$

“Affine spaces are formal duals of rings”

If X is any functor from Ring to Sets, then $\mathcal{O}_X = \text{set of all natural maps } X \rightarrow \mathbb{A}^1$. This gets a ring structure. Recall $\mathbb{A}^1 = \text{Spec}(\mathbb{Z}[t])$

$$\begin{aligned} \text{hom}(-, \text{Spec}(R)) : (\text{Aff})^{op} &\rightarrow \text{Sets} \\ \text{Spec}(S) \simeq Y &\mapsto [Y, \text{Spec}(R)] = [\text{Spec}(S), \text{Spec}(R)] \cong [R, S] \end{aligned}$$

Each ring morphism $B \rightarrow A$ defines a morphism of schemes $\text{spec}A \rightarrow \text{spec}B$ and conversely all morphisms between them arise in this fashion. We take as definition of Spec

$$\begin{aligned} \text{Spec}(R) : \text{Cring} &\rightarrow \text{Sets} \\ S &\mapsto [R, S] \end{aligned}$$

Example 1. The two common schemes that we encounter are $\mathbb{A}^1(R) = \text{Sm}_k(\text{Spec}(R), \text{Spec}(k[t])) = k - \text{alg}(k[t], R) = R$ and $\mathbb{G}_m := \text{spec} k[t, t^{-1}] = \mathbb{A}_k^1 \setminus \{0\}$

2 Motivic Spaces

We take Sm_S and embed them into the presheaf category as it is well-behaved. Zariski topology has too few covers. étale topology has enough geometry but infinite cohomological dimension. So we use Nisnevich which is

finer than Zariski but coarser than étale.

Theorem 2. A presheaf $F : \mathbf{Sm}_S^{op} \rightarrow S$ is a Nisnevich sheaf iff

- (1) $F(\Phi) \simeq *$
- (2) Let

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & \lrcorner & \downarrow p \\ U & \xrightarrow{i} & X \end{array}$$

be a Nisnevich square in $\mathbf{Sm}_S$, i.e. it is cartesian i is an open immersion, p is étale and $p^{-1}(X - U) \rightarrow X - U$ is an isomorphism. Then the square

$$\begin{array}{ccc} F(X) & \longrightarrow & F(V) \\ \downarrow & & \downarrow \\ F(U) & \longrightarrow & F(W) \end{array}$$

is a pullback square.

Also the Yoneda embedding $\mathbf{Sm}_S \rightarrow \mathbf{Pre}(\mathbf{Sm})_S$, sends Nisnevich squares to pushout squares.

Definition 3. A presheaf $F \in \mathbf{Pre}(\mathbf{Sm})_S$ is called $\mathbb{A}^1$ -invariant if for all $X \in \mathbf{Sm}_S$ the map $\mathrm{pr}^* : F(X) \rightarrow F(X \times \mathbb{A}^1)$ is an equivalence.

Definition 4. $\mathbf{Spc}S \subset \mathbf{sPre}(\mathbf{Sm})_S$ is the full sub category of $\mathbb{A}^1$ -invariant Nisnevich sheaves. This is the ∞ -category of motivic spaces. So we have

$$\mathbf{MotSpc} = L_{\mathbb{A}^1} \mathbf{Sh}_{\mathrm{Nis}}(\mathbf{Sm}_S)$$

$\mathbf{Spc}(S)$ is not an ∞ -topos.

Definition 5. A map $H : X \times \mathbb{A}^1 \rightarrow Y$ in $\mathbf{Sm}_S$ is called $\mathbb{A}^1$ -homotopy.

We write $\mathbb{G}_m$ for the pointed S -scheme $(\mathbb{A}^1 - \{0\}, 1)$.

The squares

$$\begin{array}{ccc} \mathbb{G}_m \times \mathbb{G}_m & \longrightarrow & \mathbb{G}_m \times \mathbb{A}^1 \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{A}^1 \times \mathbb{G}_m & \longrightarrow & \mathbb{A}^2 - \{0\} \end{array} \quad \begin{array}{ccc} \mathbb{G}_m & \longrightarrow & \mathbb{A}^1 \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{A}^1 & \xrightarrow{\neq} & \mathbb{P}^1 \end{array}$$

are Nisnevich squares. So we obtain pushout squares

$$\begin{array}{ccc} \mathbb{G}_m \times \mathbb{G}_m & \longrightarrow & \mathbb{G}_m \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{G}_m & \longrightarrow & \mathbb{A}^2 - \{0\} \end{array} \quad \begin{array}{ccc} \mathbb{G}_m & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \mathbb{P}^1 \end{array}$$

in $\mathbf{Spc}(S)$ after contracting $\mathbb{A}^1$. We deduce $\mathbb{A}^2 - \{0\} \simeq \Sigma(\mathbb{G}_m \wedge \mathbb{G}_m)$ and $\mathbb{P}^1 \simeq S^1 \wedge \mathbb{G}_m$ in $\mathbf{Spc}(S)_*$.

Definition 6. For integers $s \geq w \geq 0$, the motivic spheres are $S^{s,w} = S^{s-w} \wedge \mathbb{G}_m^w \in \mathbf{Spc}(S)_*$. So $\mathbb{P}^1 \simeq S^{2,1}$ and $\mathbb{A}^2 - \{0\} \simeq S^{3,2}$.

There is an equivalence $S^{2n-1,n} \simeq \mathbb{A}^n - \{0\}$.

Definition 7. The category $(\mathcal{SH}(S), \otimes)$ is the initial presentably symmetric monoidal category under $\mathbf{Spc}(S)_*$, i.e. it comes with a functor $\Sigma^\infty : \mathbf{Spc}(S)_* \rightarrow \mathcal{SH}(S)$, on which tensoring with $S^{s,w} = \Sigma^\infty S^{s,q}$ defines an equivalence $\Sigma^{s,w} = S^{s,w} \otimes - : \mathcal{SH}(S) \rightarrow \mathcal{SH}(S)$. We invert $\mathbb{P}^1 \simeq S^1 \wedge \mathbb{G}_m$

Theorem 8. Let

$$\mathrm{Spc}(S)_*[(\mathbb{P}^1)^{-1}] = \mathrm{colim} \left(\mathrm{Spc}(S)_* \xrightarrow{(-) \wedge \mathbb{P}^1} \mathrm{Spc}(S)_* \xrightarrow{(-) \wedge \mathbb{P}^1} \mathrm{Spc}(S)_* \rightarrow \cdots \right)$$

in Pr^L .

3 Realization

If X is a scheme over k and R is in $k\text{-CAlg}$. Then $X(R)$ is the set of R valued points of X is the set of k -morphisms $\mathrm{Spec}(R) \rightarrow X$. If k is a field and $R = k$, the set of k morphisms identifies with the set of points $x \in X$.

A k -valued point on X is $\underline{a} = (a_1, \dots, a_n) \in k^n$ such that $f_i(\underline{a}) = 0, \forall i$. Thus

$$\mathrm{hom}_{k\mathrm{Alg}}(A, k) \cong \mathrm{hom}_{\mathrm{Sm}_k}(\mathrm{Spec}(k), X)$$

So if X is an S -scheme and T also an S -scheme, then the set of T -points on X is $X(T) := \mathrm{hom}_S(T, X)$.

More generally, if X and Y are schemes, a morphism $Y \rightarrow X$ is a Y valued point of X . If $Y = \mathrm{Spec} A$ is affine then the set of Y valued points of X is just $\mathrm{hom}(Y, X)$ and the functor $Y \mapsto \mathrm{hom}(Y, X)$ is called the functor points of the scheme X , since to a given scheme Y it assigns the set of Y -valued points

Example 9. $\mathbb{A}^1(R) = \mathrm{Sm}_k(\mathrm{Spec}(R), \mathrm{Spec}(k[t])) = k\text{-alg}(k[t], R) = R$. This also tells us that $\mathbb{A}_k^n(R) = R^n$.

Example 10. The scheme $\mathbb{G}_m := \mathrm{spec} k[t, t^{-1}] = \mathbb{A}_k^1 \setminus \{0\}$ and $\mathbb{G}_m(R) = R^\times$.

Let $X \in \mathrm{Sm}_k$. The *real points* of X form a space, denoted by $X(\mathbb{R})$, with underlying set $\mathrm{Sm}_k(\mathrm{Spec}(\mathbb{R}), X) \simeq \mathrm{Sm}_{\mathbb{R}}(\mathrm{Spec}(\mathbb{R}), X \times_k \mathrm{Spec}(\mathbb{R}))$, and topology induced as follows: since X is of finite type over k , we may pick an open cover of X by affine subsets of the form $U = \mathrm{Spec}(k[X_1, \dots, X_n]/(f_1, \dots, f_r)) \rightarrow \mathrm{Spec}(k)$. Then, the topology on

$$\mathrm{Sm}_k(\mathrm{Spec}(\mathbb{R}), U) \simeq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid f_i(x_1, \dots, x_n) = 0 \forall i \leq r\}$$

is induced by the Euclidean one. This construction extends to a functor $(-)(\mathbb{R}) : \mathrm{Pre}(\mathrm{Sm})_S \rightarrow \mathrm{Spc}$.

Example 11. $\mathbb{R}$ maybe viewed as the C_2 -fixed points of the conjugation action on $\mathbb{C}$. Adopting an equivariant perspective allows one to compute the real realization from the complex one; or more precisely as the geometric C_2 -fixed points of some genuine equivariant spectrum. The complex realization is $r_{\mathbb{C}}(T) \simeq \mathbb{S}^1 \wedge \mathbb{C} - \{0\} \simeq \mathbb{S}^2$.

4 Presheaves with transfer

Let X, Y be algebraic varieties over F a field and assume that X is smooth. An elementary multi-valued map is an irreducible subvariety $W \subset X \times Y$ such that $W \rightarrow X$ is finite and surjective. A multi-valued map is a $\mathbb{Z}$ -linear combination of multi-valued maps. The group of multi-valued maps is denoted by $\mathrm{Cor}(X, Y)$. Cor_F is the category where objects are smooth algebraic varieties and morphisms are multi-valued maps. A presheaf with transfer is a contravariant additive functor $\mathcal{F} : \mathrm{Cor}_F \rightarrow \mathrm{Ab}$. $\mathbb{Z}_{\mathrm{tr}}(X)(U) = \mathrm{Cor}(U, X)$. We have $\mathbb{Z}_{\mathrm{tr}}(\mathrm{Spec}_F) = \mathbb{Z}$.

With this in mind we'll define the group $\mathrm{Cor}(X, Y)$, which will be our morphisms between X, Y in the category Cor_k , to be the what we would get if we allowed multivalued functions between schemes. That is, if a function defines a graph which is an isomorphism with the first factor under the projection, a correspondence will define a subscheme of the product which is finite and surjective under the projection to the first factor. There is a faithful embedding $\mathrm{Sm}_k \hookrightarrow \mathrm{Cor}_k$

Motivic cohomology admits more functoriality than just the contravariant functoriality of schemes, which we encode by $\mathrm{Cor}(S)$. This is reminiscent of the wrong-way maps in classical algebraic topology, called *transfers*. For any map of spaces $X \rightarrow Y$ with finite and discrete homotopy fibers we get a map, Becker-Gottlieb transfer,

$\Sigma_+^\infty Y \rightarrow \Sigma_+^\infty X$, which sends a point $y \in Y$ to the sum of points of the fiber. We import these to the motivic setting using the notion of correspondence of spaces.

Any map $f : X \rightarrow Y$ in $\mathbf{Sm}_S$ admits a graph Γ_f , defining a functor $\Gamma : \mathbf{Sm}_S \rightarrow \mathbf{Cor}(S)$. $\mathbf{Cor}$ is additive and symmetric monoidal. Γ induces by Yoneda extension a functor $\mathbf{sPre}(\mathbf{Sm})_S \rightarrow \mathbf{Pre}(\mathbf{Cor})(S)$ which is an algebra through Day convolution. With this we can define $\mathbf{Spc}_{\mathrm{tr}}(S) = \mathbf{Pre}^\Sigma(\mathbf{Cor}(S)) \otimes_{\mathbf{sPre}(\mathbf{Sm}_S)} \mathbf{Spc}_*(S)$.

Let $\mathbf{Spc}_{\mathrm{tr}}(S) \subset \mathbf{Pre}(\mathbf{Cor})_S$ with Nisnevich sheaves and $\mathbb{A}^1$ -invariance be defined as the category of sheaves with transfers on $\mathbf{Cor}_S$. There is a forgetful functor

$$U : \mathbf{Pre}^{\mathrm{tr}}(\mathbf{Cor})_S \rightarrow \mathbf{sPre}(\mathbf{Sm}_S)$$

This has a left adjoint called the transfer functor denoted by $\mathbb{Z}^{\mathrm{tr}}$ or $R^{\mathrm{tr} \ 1}$. We define the stable homotopy category

$$SH_{\mathrm{tr}}(S, R) = \mathbf{Spc}_{\mathrm{tr}}(S)[(\mathbb{Z}^{\mathrm{tr}} \mathbb{P}^1)^{-1}] \simeq \mathbf{Spc}_{\mathrm{tr}}(S) \otimes_{\mathbf{Spc}_*(S)} SH(S)$$

Putting this all together, the motivic picture looks like this:

$$\begin{array}{ccccccc}
\mathbf{Sm}_S & \longrightarrow & \mathbf{Pre}(\mathbf{Sm})_S & \longrightarrow & \mathbf{Spc}(S) & \xrightleftharpoons[\Omega_{\mathbb{P}^1}^\infty]{\Sigma_{\mathbb{P}^1}^\infty} & SH(S) \\
\downarrow & & \uparrow U \quad \downarrow \mathbb{Z}^{\mathrm{tr}} & & \uparrow U \quad \downarrow \mathbb{Z}^{\mathrm{tr}} & & \uparrow U \quad \downarrow \mathbb{Z}^{\mathrm{tr}} \\
\mathbf{Cor}_S & \longrightarrow & \mathbf{Pre}(\mathbf{Cor})_S & \longrightarrow & \mathbf{Spc}_{\mathrm{tr}}(S) & \xrightleftharpoons[\Omega_{\mathbb{P}^1}^\infty]{\Sigma_{\mathbb{P}^1}^\infty} & SH_{\mathrm{tr}}(S)
\end{array}
\begin{array}{l}
\swarrow H\mathbb{Z} \wedge - \\
\searrow H\mathbb{Z} \\
\text{Mod}_{H\mathbb{Z}}
\end{array}$$

5 Eilenberg-MacLane Spectrum, Motivic Cohomology

Let $HR = U_{\mathrm{tr}}(\mathbb{1}_{\mathcal{SH}_{\mathrm{tr}}(S, R)})$. We have an adjunction

$$\mathbf{Mod}_{HR} \xrightleftharpoons[\Psi]{\quad} \mathbf{Mod}_{\mathbb{1}}(\mathcal{SH}_{\mathrm{tr}}(S, R)) = \mathcal{SH}_{\mathrm{tr}}(S, R)$$

The naive EM spectrum represents cohomology but doesn't work for Dold-Thom!

Let A be an R -module and $X \in \mathbf{Sm}_S$ be essentially smooth over a field k . Then,

$$H^{p,q}(X, A) \cong [\Sigma_+^\infty X, \Sigma^{p,q} HA]$$

The fancier version using transfers

$$H^{p,q}(X; \mathbb{Z}) = H_{\mathrm{Zar}}^{p-q}(Z, \mathbb{Z}_{\mathrm{tr}}(\mathbb{G}_m^{\otimes q})[-q])$$

References

Notes by Qi Zhu

Presheaves with transfer

Motivic Cohomology

<https://mathoverflow.net/questions/259894/why-presheaves-with-transfer>

Motivic cohomology

On the real realization of the motivic spectrum ko

¹It should all be $\mathbf{Cor}(S, R)$ for correspondences of schemes over S with values taken in R -modules. If we let it take values in Ab the $R = \mathbb{Z}$. But I dropped R everywhere.